

Name: \_\_\_\_\_

# FRACTIONS, DECIMALS AND PERCENTAGES (CONVERTING BETWEEN FDP)

Date: \_\_\_\_\_

**LO: I can convert fluently between fractions, decimals and percentages.**

## What does converting mean?

**Converting** means writing a number in a **DIFFERENT** form without changing its value. Fractions, decimals and percentages are three ways to write the same amount.

**Key fact:** To convert a fraction to a decimal, divide the numerator by the denominator. To convert a decimal to a percentage, multiply by 100.

## Equivalent values

|   |                             |                          |
|---|-----------------------------|--------------------------|
| ● | $\frac{1}{2} = 0.5 = 50\%$  | all equivalent           |
| ● | $\frac{1}{4} = 0.25 = 25\%$ | all equivalent           |
| ● | $\frac{3}{5} = 0.6 = 60\%$  | all equivalent           |
| ● | $\frac{1}{3} = 0.3 = 30\%$  | $0.3 \neq \frac{1}{3}$   |
| ● | $0.7 = 70\% = \frac{7}{10}$ | should be $\frac{7}{10}$ |

## Key idea

Fraction    Decimal: divide top by bottom. Decimal  
Percentage:  $\times 100$ . Percentage    Fraction: write over 100  
and simplify.

$$\frac{3}{8}$$

$$3 \div 8 = 0.375$$

$$37.5\%$$

Divide, then multiply by 100

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - fraction to decimal

Convert  $\frac{3}{4}$  to a decimal

$$3 \div 4 \\ = 0.75$$

$$= 0.75$$

Divide numerator by denominator:  $3 \div 4 = 0.75$

### Example 2 - decimal to percentage

Convert 0.35 to a percentage

$$0.35 \times 100 \\ = 35 \\ \text{Add \% sign}$$

$$= 35\%$$

Multiply the decimal by 100 and add the % symbol.

### Example 3 - percentage to fraction

Convert 45% to a fraction in simplest form

$$45/100 \\ \text{HCF of 45 and 100} = 5 \\ = 9/20$$

$$= \frac{9}{20}$$

Write over 100, then simplify by dividing both by the HCF (5).

### Example 4 - fraction to percentage

Convert  $\frac{7}{20}$  to a percentage

$$7 \div 20 = 0.35 \\ 0.35 \times 100 \\ = 35\%$$

$$= 35\%$$

Convert to decimal first (divide), then multiply by 100.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Fraction to decimal

Convert each fraction to a decimal by dividing the numerator by the denominator.

- a)  $\frac{1}{2}$
- b)  $\frac{1}{5}$
- c)  $\frac{3}{8}$
- d)  $\frac{7}{20}$

### 2. Decimal to percentage

Convert each decimal to a percentage by multiplying by 100.

- a) 0.4
- b) 0.75
- c) 0.08
- d) 0.125

### 3. Percentage to fraction

Convert each percentage to a fraction in its simplest form. Write over 100 then simplify.

- a) 20%
- b) 60%
- c) 35%
- d) 84%

### 4. Fraction to percentage

Convert each fraction to a percentage. Divide first, then multiply by 100.

- a)  $\frac{1}{4}$
- b)  $\frac{2}{5}$
- c)  $\frac{3}{8}$
- d)  $\frac{11}{20}$

### 5. Mixed conversions

Convert each value to the other two forms (fraction, decimal, percentage).

- a)  $0.6 = \quad = \quad \%$       b)  $72\% = \quad = \quad$       c)  $\frac{9}{25} = \quad = \quad \%$       d)  $0.875 = \quad = \quad \%$       e)  $5\% = \quad = \quad$

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a)  $\frac{1}{3} = 0.3$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b)  $0.25 = 25\%$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c)  $70\% = \frac{7}{10}$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d)  $0.5\% = \frac{1}{2}$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e)  $\frac{4}{5} = 0.45$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop offers a  $\frac{1}{4}$  off sale. The sale sign says "20% off everything!"  
Is the sale sign correct? Show your working to explain why or why not.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) Three students scored the following on different tests:  
Amy: 17 out of 20   Ben: 0.82   Clara: 78%  
Who got the highest score? Convert all to the same form to compare.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

